Uhlmann's Theorem, Talk 4 July 16/26 Ø

Summary of the Relevant Properties of Fidelity

We use the **squared-fidelity convention:**

$$F(\rho,\sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1^2,$$

with root fidelity

$$\sqrt{F(\rho,\sigma)} = \|\sqrt{\rho}\sqrt{\sigma}\|_1.$$

Versus trace distance:

$$T(\rho,\sigma) = \frac{1}{2}\|\rho-\sigma\|_1,$$

1. Boundedness

$$0 \le F(\rho,\sigma) \le 1.$$

Fidelity is a normalized **measure of similarity**, with larger values representing greater overlap.

2. Equality condition

$$F(\rho,\sigma) = 1 \quad \Leftrightarrow \quad \rho = \sigma.$$

Two normalized quantum states have maximum fidelity exactly when they are the same state.

3. Orthogonality condition

$$F(\rho,\sigma) = 0 \quad \Leftrightarrow \quad \mathrm{supp}(\rho) \perp \mathrm{supp}(\sigma).$$

Fidelity vanishes exactly when the states have orthogonal supports* and are therefore perfectly distinguishable.

*$\mathrm{supp}(\rho) = (\ker\rho)^{\perp}$ is the linear subspace of the domain of ρ that maps to non-zero vectors. It's dimension is the rank of ρ.

4. Symmetry

$$F(\rho,\sigma) = F(\sigma,\rho).$$

Fidelity does not depend on the order in which the two states are compared. fidelity_and_gentle_measurement...

5. ✅ Pure-state formula

For pure states,

$$F(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = |\langle\psi|\phi\rangle|^2.$$

Fidelity reduces to the ordinary squared overlap, or transition probability, between the two state vectors.

6. ✅ Pure-mixed formula

If one state is pure, then

$$F(|\psi\rangle\langle\psi|, \rho) = \langle\psi|\rho|\psi\rangle.$$

This is the probability that ρ passes the projective test for being the target state $|\psi\rangle$.

7. ✅ Commuting or classical case

If

$$\rho = \sum_x p_x |x\rangle\langle x|, \qquad \sigma = \sum_x q_x |x\rangle\langle x|,$$

then

$$F(\rho,\sigma) = \left(\sum_x \sqrt{p_x q_x}\right)^2.$$

Quantum fidelity reduces to the square of the classical Bhattacharyya overlap when the states commute. fidelity_and_gentle_measurement...

8. 📌 Uhlmann's theorem

$$F(\rho,\sigma) = \max_{\substack{|\psi_\rho\rangle, |\psi_\sigma\rangle \\ \text{purifications of } \rho,\sigma}} |\langle\psi_\rho|\psi_\sigma\rangle|^2.$$

Fidelity is the largest possible squared overlap between purifications of the two mixed states. fidelity_and_gentle_measurement...

9. Unitary and isometric invariance

For every unitary or isometry V,

$$F(V\rho V^\dagger, V\sigma V^\dagger) = F(\rho,\sigma).$$

Reversible embeddings and unitary evolution preserve fidelity exactly. talk_3_outline(8)

10. Multiplicativity under tensor products

$$F(\rho_1 \otimes \rho_2, \sigma_1 \otimes \sigma_2) = F(\rho_1,\sigma_1)\,F(\rho_2,\sigma_2).$$

The fidelity of product states is the product of the fidelities of their individual components. fidelity_and_gentle_measurement...

11. 📌 Monotonicity under quantum channels

For every quantum channel $\mathcal{N}$,

$$F(\mathcal{N}(\rho), \mathcal{N}(\sigma)) \geq F(\rho, \sigma).$$

Applying the same physical process to both states cannot decrease their fidelity; noise can make states more similar. fidelity_and_gentle_measurement...

12. Monotonicity under partial trace

For bipartite states,

$$F(\rho_A, \sigma_A) \geq F(\rho_{AB}, \sigma_{AB}),$$

where

$$\rho_A = \mathrm{Tr}_B\, \rho_{AB}, \qquad \sigma_A = \mathrm{Tr}_B\, \sigma_{AB}.$$

Discarding a subsystem cannot decrease fidelity because it removes potentially distinguishing information.

13. Behaviour under measurements

For a POVM $\{\Lambda_y\}_y$, define

$$p_y = \mathrm{Tr}(\Lambda_y \rho), \qquad q_y = \mathrm{Tr}(\Lambda_y \sigma).$$

Then

$$\sum_y \sqrt{p_y q_y} \geq \sqrt{F(\rho, \sigma)}.$$

Every measurement produces classical distributions whose overlap is at least the root fidelity of the original quantum states. talk_3_outline(8)

14. ‼ Fuchs–van de Graaf inequalities

With trace distance

$$T(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1,$$

we have

$$1 - \sqrt{F(\rho, \sigma)} \leq T(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)}.$$

Fidelity and trace distance define compatible notions of closeness and allow estimates in one measure to be converted into estimates in the other. fidelity_and_gentle_measurement...

15. 📌 Exact pure-state relation with trace distance

For pure states,

$$T(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = \sqrt{1 - F(\psi, \phi)}.$$

For pure states, fidelity and trace distance determine one another exactly.

16. 📌 Purified distance

$$P(\rho, \sigma) = \sqrt{1 - F(\rho, \sigma)}.$$

Purified distance converts fidelity into a genuine distance that behaves especially well with purifications and approximate quantum states.

17. 📌 Contraction of purified distance under channels

$$P(\mathcal{N}(\rho), \mathcal{N}(\sigma)) \leq P(\rho, \sigma).$$

Physical processing cannot increase purified distance, just as it cannot increase trace distance. fidelity_and_gentle_measurement...

18. Fidelity itself is not a metric

Fidelity is a similarity measure rather than a distance: identical states have fidelity 1, and fidelity itself does not satisfy the usual metric axioms such as the triangle inequality.

UHLMANN'S THEOREM

Properties of Fidelity

- see pdf summary sheets.

Uhlmann's Thm

We have the defn of (squared) fidelity btwn two states:

$$F(\rho,\sigma) = \| \sqrt{\rho}\sqrt{\sigma} \|_1^2 \quad \text{for states on } A$$

Then also,

$$F(\rho_A, \sigma_A) = \max_{\text{purifications}} |\langle \psi_\rho | \psi_\sigma \rangle|^2$$

where the maximum is over all purifications,

$|\psi_\rho\rangle_{RA}$ of ρ_A

$|\psi_\sigma\rangle_{RA}$ of σ_A,

with an extended reference system $R \otimes A$ & $\dim R = \dim A$.

Side Defn's & Identities First:

① Defn of a maximally entangled state, $|\Gamma\rangle$

② Identity: $\langle \Gamma | M_R \otimes N_A | \Gamma \rangle = \mathrm{tr}(M^T N)$

③ Using $|\Gamma\rangle$ to purify states.

④ Alternative trace 1-norm defn: $\|X\|_1 = \max_{\substack{V \text{ unitary} \\ (V^\dagger V = I)}} |\mathrm{tr}\, VX|$

① **Defn:**

A maximally entangled state:

$$|\Gamma\rangle_{RA} \equiv \sum_{j=1}^{d} |j\rangle_R |j\rangle_A \quad \text{on } R \otimes A.$$

where $\{|j\rangle_R\}_{j=1}^{d}$, $\{|j\rangle_A\}_{j=1}^{d}$ are arbitrary ONB for R & A, resp'y.

• note we will use the unormalized version above as an algebraic tool → properly we should divide by $\sqrt{d}$.

Why "maximally entangled"?

Take normalized version,

$$|\Gamma\rangle'_{RA} = \frac{1}{\sqrt{d}} \sum_j |j\rangle_R |j\rangle_A \implies$$ Compare to $d=2$ Bell State $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Then,

$$|\Gamma\rangle'_{RA}\langle\Gamma|'_{RA} = \frac{1}{d}\sum_{ij} |j\rangle|j\rangle\langle i|\langle i|$$

$$= \frac{1}{d}\sum_{ij} (|j\rangle_R \otimes |j\rangle_A)(\langle i|_R \otimes \langle i|_A)$$

$$= \frac{1}{d}\sum_{ij} |j\rangle\langle i|_R \otimes |j\rangle\langle i|_A .$$

Now "trace out" system R: $tr_R = tr \otimes I_A$ ← on subsystem R; ← leave subsystem A alone.

$$(tr \otimes I)|\Gamma\rangle'\langle\Gamma|' = \frac{1}{d}\sum_{ij} tr|j\rangle\langle i|_R \otimes |j\rangle\langle i|_A$$

$$= \frac{1}{d}\sum_{ij} \delta_{ij} \cdot |j\rangle\langle i|_A$$

$$= \frac{1}{d}\sum_{i} |i\rangle\langle i|_A$$

$$= \frac{1}{d} I_A \implies \text{similarly, } tr_A|\Gamma\rangle'\langle\Gamma|' = \frac{1}{d} I_R.$$

→ so each subsystem in isolation is maximally (completely) mixed a a single particle state.

This is the definition of maximal entanglement between two systems.

② Identity: $\langle \Gamma | M_R \otimes N_A | \Gamma \rangle = \mathrm{tr}\, M^T N$ ← Note: ~~Not~~ $\dagger$ (adjoint).

a) Transpose Trick for $|\Gamma\rangle$.

$$\boxed{(M \otimes I)|\Gamma\rangle = (I \otimes M^T)|\Gamma\rangle}$$

Note: $A|j\rangle = \sum_i A_{ij} |i\rangle$ $\xrightarrow{\text{Check}}$ $\langle k|A|j\rangle \overset{?}{=} \sum_i A_{ij} \langle k|j\rangle$

$= \sum_i A_{ij} \delta_{kj}$

$= A_{kj}$ ✓

Do LHS = RHS on T.T:

$$\text{LHS} = (M \otimes I) \sum_j |j\rangle \otimes |j\rangle$$

$$= \sum_j M|j\rangle \otimes |j\rangle$$

$$= \sum_{ji} M_{ij} |i\rangle \otimes |j\rangle$$

$$\text{RHS} = (I \otimes M^T) \sum_j |j\rangle \otimes |j\rangle$$

$$= \sum_j |j\rangle \otimes M^T |j\rangle$$

$$= \sum_j |j\rangle \otimes \sum_i [M^T]_{ij} |i\rangle$$

$$= \sum_j |j\rangle \otimes M_{ji} |i\rangle$$

$$= \sum_{ji} M_{ji} |j\rangle \otimes |i\rangle$$

Swap $i \leftrightarrow j$ indices.

∴ LHS = RHS.

b) Now use T.T. to show $\langle \Gamma | M_R \otimes N_A | \Gamma \rangle = \mathrm{tr}\, M^T N$

pg ④.

②b) Cont'd.

We have the transpose trick: $(M\otimes I)|\Gamma\rangle = (I\otimes M^T)|\Gamma\rangle$

We want to show: $\langle\Gamma|(M\otimes N)|\Gamma\rangle = \mathrm{tr}\, M^T N$.

So, $\boxed{\langle\Gamma|M\otimes N|\Gamma\rangle} = \langle\Gamma|(M\otimes I)(I\otimes N)|\Gamma\rangle$

$= \langle\Gamma|(I\otimes M^T)(I\otimes N)|\Gamma\rangle$

$\boxed{= \langle\Gamma|(I\otimes M^T N)|\Gamma\rangle}$. (*)

But in general,

$$\langle\Gamma|I\otimes X|\Gamma\rangle = \sum_i\sum_j \overbrace{\langle i|\otimes\langle i|\,(I\otimes X)}^{\text{in } R}\,\underbrace{|j\rangle\otimes|j\rangle}_{\text{in } A}$$

$$= \sum_{ij}\underbrace{\langle i|j\rangle}_{\delta_{ij}}\otimes\langle i|X|j\rangle$$

$$= \sum_i \langle i|X|i\rangle$$

$$= \sum_i X_{ii}$$

$$\boxed{\langle\Gamma|I\otimes X|\Gamma\rangle = \mathrm{tr}\, X}.$$

So from (*) above,

$$\langle\Gamma|M\otimes N|\Gamma\rangle = \langle\Gamma|I\otimes M^T N|\Gamma\rangle$$

$$\therefore\ \boxed{\langle\Gamma|M\otimes N|\Gamma\rangle = \mathrm{tr}\, M^T N}$$

③ Using $|\Gamma\rangle$ to purify states:

- recall previous method to purify a ρ_A:

$$\rho_A \longrightarrow |\psi_\rho\rangle_{RA} = \sum_i \underbrace{\sqrt{p_i}}_{\text{eigenvalues of } \rho_A} \underbrace{|i\rangle_R}_{\text{arbitrary ONB of } R} \otimes \underbrace{|e_i\rangle_A}_{\text{ONB of eigenvectors of } \rho_A}$$

- alternative : $|\psi_\rho\rangle_{RA} = (I_R \otimes \sqrt{\rho_A})|\Gamma\rangle_{RA}$

is <u>also</u> a purification of ρ_A.

- check:

$$|\psi_\rho\rangle_{RA}\langle\psi_\rho|_{RA} = (I_R \otimes \sqrt{\rho_A})|\Gamma\rangle\langle\Gamma|(I_R \otimes \sqrt{\rho_A}).$$

$$= (I \otimes \sqrt{\rho}) \sum_{ij} |i\rangle|i\rangle\langle j|\langle j| \, (I_R \otimes \sqrt{\rho}).$$

$$= (I \otimes \sqrt{\rho}) \sum_{ij} |i\rangle\langle j|_R \otimes |i\rangle\langle j|_A \, (I \otimes \sqrt{\rho}).$$

$$= \sum_{ij} |i\rangle\langle j|_R \otimes \sqrt{\rho_A}|i\rangle\langle j|_A \sqrt{\rho_A}$$

- $$(\mathrm{tr} \otimes I_A)|\psi_\rho\rangle\langle\psi_\rho| = \sum_{ij} \underbrace{\mathrm{tr}\,|i\rangle\langle j|}_{\delta_{ij}} \otimes \sqrt{\rho_A}|i\rangle\langle j|_A \sqrt{\rho_A}$$

$$= \sum_i 1 \otimes \sqrt{\rho_A}|i\rangle\langle i|\sqrt{\rho_A}$$

$$= \sqrt{\rho_A} \underbrace{\sum_i |i\rangle\langle i|}_{I_A} \sqrt{\rho_A}$$

$$= \rho_A \checkmark$$

(4) Alternative trace 1-norm definition.

We have: $\|X\|_1 = \max\limits_{\substack{\text{unitary } U \\ U^\dagger U = I}} |\operatorname{tr} UX|$

- think of as a generalization of scalar case for $\mathbb{C}$-numbers.
  - for $x \in \mathbb{C}$, $u \in \mathbb{C}$ on unit circle ($e^{i\theta}$).

    what is $\max\limits_{u = e^{i\theta}} |ux|$?

    $\Rightarrow |ux| = |u| \cdot |x|$

    $= |x| \quad \because |u| = 1 \;\; \forall\, u \in \{e^{i\theta}\}$.

    $\therefore \max\limits_{u=e^{i\theta}} |ux| = |x|$ trivially.
- can show for general operators on $\mathbb{C}^n$.

We now have the pieces necessary to prove Uhlmann's Theorem.

$\Rightarrow$ p. (7).

Uhlmann's Theorem:

Given the fidelity of two states ρ, σ defined as,

$$F(\rho,\sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1^2 \quad \text{for states } \rho,\sigma \text{ on } A.$$

Then, $$F(\rho_A, \sigma_A) = \max_{\text{purifications}} \left|\langle \psi_{\rho_A} | \psi_{\sigma_A}\rangle\right|^2$$

where $$\rho_A \xrightarrow{\text{purif.}} |\psi_{\rho_A}\rangle_{RA}$$

$$\sigma_A \xrightarrow{\text{purif.}} |\psi_{\sigma_A}\rangle_{RA}$$

PROOF:

① Choose the $|\Gamma\rangle$ Purification.

- choose (for now) arbitrary ONB $\{|i\rangle_A\}$ for system A and $\{|i\rangle_R\}$ for reference system R with $\dim R = \dim A$.
- define the maximally mixed state.

$$|\Gamma\rangle_{RA} = \sum_i |i\rangle_R |i\rangle_A .$$

- define the purifications:

$$\left.\begin{aligned} |\psi_\rho\rangle_{RA} &= (I_R \otimes \sqrt{\rho})|\Gamma\rangle_{RA} \\ |\psi_\sigma\rangle_{RA} &= (I_R \otimes \sqrt{\sigma})|\Gamma\rangle_{RA} \end{aligned}\right\} (*)$$

⑧

② Reduce max^{m}'s over purif^{n}'s to max^{n} over unitaries.

- recall all possible purifications of a given state are related by unitary transformations on the reference system:

All possible purifications are given by:

(*)
$$\left.\begin{array}{l} (U_R^1 \otimes I_A)|\psi_\rho\rangle_{RA} \\ \\ (U_R^2 \otimes I_A)|\psi_\sigma\rangle_{RA} \end{array}\right\}$$

- U_R^1, U_R^2 are unitary on R
- all purf^{n}s can be had by varying U_R^1, U_R^2 over $\mathcal{U}$ (all unitary operators).

- so, all possible overlaps of purifications are:

$$\langle\psi_\rho|(U_R^{1\dagger} \otimes I_A)(U_R^2 \otimes I_A)|\psi_\sigma\rangle$$

$$= \langle\psi_\rho|(U_R^{1\dagger} U_R^2 \otimes I_A)|\psi_\sigma\rangle$$

$$= \langle\psi_\rho|U_R \otimes I_A|\psi_\sigma\rangle \quad \text{where } U_R \equiv U_R^{1\dagger} U_R^2 \text{ is also unitary.}$$

$\Rightarrow$ we only have to vary the single U_R to get all overlaps.

- so we can write:

$$\boxed{\max_{\text{purifications}} |\langle\psi_\rho|\psi_\sigma\rangle| = \max_{U_R \in \mathcal{U}} |\langle\psi_\rho|(U_R \otimes I_A)|\psi_\sigma\rangle|}$$